ZEESHAN HAIDER

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SUMMARY

Mechatronics Engineering student specializing in **robotics, aerospace systems, UAVs, and AI-based computer vision**. Experienced in **YOLOv11, PyTorch, OpenCV, ONNX, INT8 quantization, MATLAB/Simulink, SolidWorks, and ANSYS**, with hands-on experience in UAV detection, model optimization, robotic design, and simulation. Research experience includes **bio-inspired UAV wing design, CFD/FEA analysis, fabrication, and experimental testing**.

EDUCATION AND QUALIFICATIONS

NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY (NUST), BE MECHATRONICS ENGINEERING, PAKISTAN **2023- PRESENT**

- CGPA – 3.28

WORK EXPERIENCE

AI INTERN, BAYKAR TECHNOLOGIES | RAWALPINDI **JULY 2026-AUG 2026**

- Trained and evaluated **YOLO11** on a **50,000-image UAV detection dataset**, optimizing the model for reliable real-time object detection.
- Applied **INT8 quantization** and **ONNX conversion** to optimize the trained model for efficient inference and deployment.
- Conducted comparative **benchmarking of FP32, INT8, and ONNX models**, analyzing accuracy, inference speed, model size, and computational performance.
- Performed comprehensive **model validation and performance evaluation** using detection metrics and inference benchmarks to assess the impact of optimization techniques.
- Tested the optimized UAV detection pipeline on **MP4 video**, validating end-to-end object detection performance under practical video inference conditions.

RESEARCH INTERN, NATIONAL CENTRE OF ROBOTICS AND AUTOMATION, RAWALPINDI **JUN *2025-AUG 2025***

- **Designed a bio-inspired UAV wing** mimicking a bird's flapping mechanism.
- Conducted **CFD simulations in Ansys Fluent** for aerodynamic flow, lift, and drag.
- Applied **FEA in Ansys Mechanical** for load and structural analysis.
- **Fabricated the flapping wing mechanism** using 3D printing techniques.
- Performed **wind tunnel testing** to validate aerodynamic performance.
- **Co-authored** a research paper, contributing to design, analysis, and documentation (under review).
- **Industrial Design Patent** submission to the Intellectual Property Organization.

STUDENT INTERN, AL-AKBARI HOUSES BUILDERS & DEVELOPERS | REMOTE **JUL 2025-AUG 2025**

- Completed a **remote internship** with Al-Akbari Builders, contributing to mechanical engineering design tasks.
- Developed and modified **3D models in SolidWorks** for mechanical components and assemblies.
- Created and refined **2D technical drawings in AutoCAD** for design documentation and project use.

SKILLS

AI / MACHINE LEARNING

- Python (NumPy, Pandas, Matplotlib)
- PyTorch
- YOLO11 / Ultralytics
- OpenCV
- Computer Vision
- Object Detection & Tracking
- ByteTrack & Kalman Filter
- INT8 Quantization
- ONNX / ONNX Runtime
- Model Validation & Benchmarking

ROBOTICS & CONTROL

- MATLAB / Simulink
- Robotics & Robot Kinematics
- PID Control
- Arduino / ESP32 Programming • C++ (OOP, Data Structures)

RESEARCH & DEVELOPMENT TOOLS

- Technical Research & Literature Review
- LaTeX (Technical Documentation & Research Papers)
- Mendeley (Reference Management)
- Git / GitHub

ENGINEERING & DESIGN

- SolidWorks (3D Modeling, Robot Design, Design for 3D Printing)
- ANSYS Fluent & Mechanical (CFD, FEA)
- AutoCAD
- Altium Designer (PCB Design)
- LTspice (Circuit Analysis & Simulation)
- Proteus (Circuit Design & Simulation)
- Assembly Language

PROJECTS

- **BIO-INSPIRED FLAPPING BIRD MECHANISM (2025):** Designed, simulated, and fabricated bio-inspired UAV mechanisms, followed by product testing to validate performance.
- **COMPONENTSCAN - AI-BASED ELECTRONIC COMPONENT IDENTIFICATION (2026):** Developed a camera-based AI image classification system prototype integrating Arduino/ESP32, LCD, and LED feedback for automated electronic component recognition.
- **PID-BASED ROBOTIC ARM JOINT CONTROL (2026):** Developed a DC motor-driven robotic arm model and designed a MATLAB-based PID controller for payload disturbance rejection, achieving 3.75° peak deviation and 0.886s recovery time.
- **INDIGENOUS ROBOT (2025–2026):** Developed a line-following robot using QTR-MD 16RC, utilizing PID control for precision and stability, and implemented block placement.
- **SELF-BALANCING ROBOT (2026):** Designed an Arduino Uno and MPU-6050 two-wheeled robot balanced by a 200 Hz PID controller with complementary-filter sensor fusion and live serial tuning.
- **ELECTRIC BIKE MODELING & SIMULATION (2026):** Developed a MATLAB/Simulink-based model of an e-bike, incorporating vehicle dynamics, battery modeling, and field-oriented control (FOC), and analyzed system performance under varying terrain conditions.
- **EPROM BURNER (2025):** Designed and programmed an EPROM burner using AT89C51, integrating embedded systems and Assembly programming.
- **IoT-BASED SpO_2 MONITOR (2025):** Developed an IoT-based SpO_2 monitoring device for real-time oxygen saturation tracking and wireless data transmission.
- **HAND-CRANK GENERATOR (2025):** Designed and fabricated a manually operated generator using SolidWorks and 3D-printed PLA parts, integrating neodymium magnets and copper coils to demonstrate electromagnetic induction.
- **SERVO-CONTROLLED ROBOTIC ARM (2025):** Developed a servo-actuated robotic arm with y-axis motion and a gear-driven gripper, controlled via ESP32, applying mechanics of materials principles for load optimization.
- **HOME AUTOMATION SYSTEM (2024):** Created a GUI-based system in C++, applying data structures and OOP concepts to automate home appliances, including lights, fans, doors, and a thermostat.
- **LOGIC-BASED LINE FOLLOWER (2024):** Designed a high-precision, microcontroller-free robot using logic gates, QTR-8A sensors, comparators, and counter ICs for junction counting.

HOBBIES

- Sketching
- Badminton
- Cooking
- Horse Riding
- Psychology